

Chapter 20: Antepartum Fetal Assessment — Revision Notes

Williams Obstetrics 26e, pp. 978–1012 of the PDF. High-yield numbers are in **bold**. The five tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 10 figures are included from the PDF.

Big picture: Antepartum surveillance uses fetal **heart rate, movement, breathing and amniotic fluid** to prevent fetal death without unnecessary intervention. A normal test is **highly reassuring: negative predictive value $\geq 99.8\%$** , but **positive predictive value only 10–40%**. Benefit rests on circumstantial evidence — no definitive RCTs (ethical reasons).

1. Fetal Movements

Fetal behavioral states

- Fetal activity starts as early as **7 wk**. **20–30 wk**: movements organize into rest–activity cycles. **~36 wk**: true behavioral states.
- Nijhuis states:**

State	Description	Neonatal equivalent
1F	Quiescent; narrow FHR oscillation	Quiet sleep
2F	Frequent gross body movements, continuous eye movements, wider FHR oscillation	Active (REM) sleep
3F	Continuous eye movements, no body movements, no accelerations — rare, disputed	—
4F	Vigorous body movements, continuous eye movements, FHR accelerations	Awake

- From **28–30 wk** most time is in **1F and 2F**; at **38 wk**, **75%** of time.
- Bladder volume $\uparrow$ in 1F and $\downarrow$ in 2F (less urine production — reduced renal flow in active sleep).

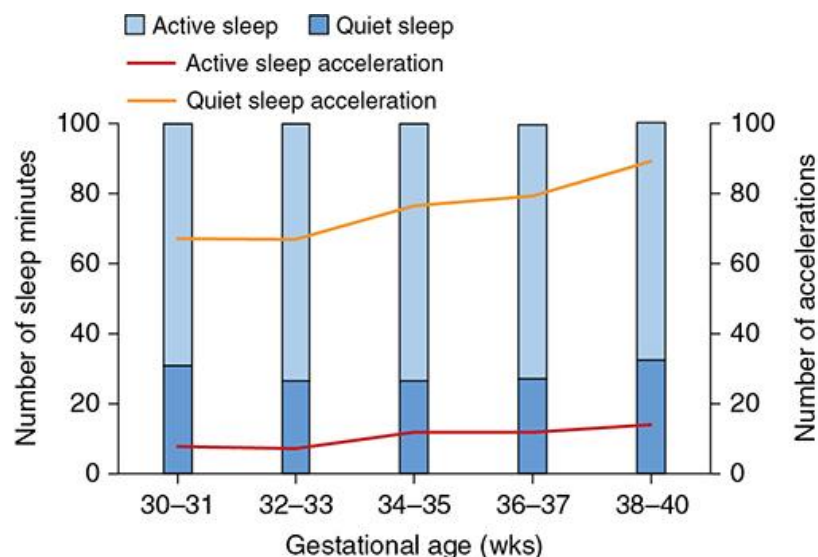


Figure 20-1. Across the 3rd trimester fetuses spend more time in active sleep and have more accelerations per hour.

Determinants of fetal activity

- Sleep–wake cycles** (independent of the mother): full cycle **~60 min** (quiet **23 min**, active **40 min**).
- ⚠ Longest normal inactivity: 75 min** (24-h scans of near-term fetuses).

- ↓ **Amnionic fluid** → ↓ movement (restricted space).
- ↓ **Movement: maternal smoking; methadone, buprenorphine; betamethasone for 24–72 h** (diurnal pattern lost; dexamethasone effect less clear).
- **Acetaminophen: no effect. Glucose load ↑ activity.**

Maternal perception

- Movements: weak, strong, rolling. Weak ones fall and vigorous ones rise as pregnancy advances, then subside at term (less fluid/space).
- **Weekly counts (12-h daily recordings): ~200 at 20 wk → peak 575 at 32 wk → 282 at 40 wk. >90% of women feel stronger movements in the evening/night.**
- Women perceive only **16–45%** of sonographically seen movements. **BMI does not reduce perception**; effect of nulliparity/anterior placenta unclear.

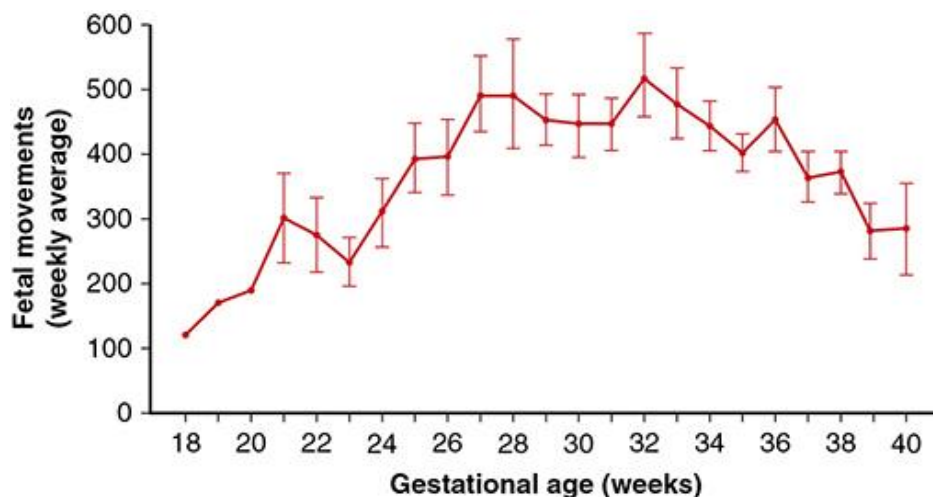


Figure 20-2. Weekly fetal movements rise to a peak at ~32 wk and decline toward term.

The text gives a peak of 575 movements/wk at 32 wk; the plotted mean at 32 wk is nearer 520.

Clinical application

- Optimal protocol undefined. Quantitative (e.g., **10 movements in 2 h**) vs subjective impression.
- **Grant (1989), >68,000 women (28–32 wk):** formal **count-to-10** (mean **2.7 h/d**) vs informal questioning → **same antepartum death rates**; most fetuses were already dead when mothers presented → **informal impression is as valid as formal counting**.
- **Decreased fetal movement in 6–7%** of pregnancies, **without ↑ stillbirth**; a 400,000-pregnancy education + management package **did not reduce stillbirth**.
- May help detect **FGR**; counting ↓ **1-min Apgar ≤3 (0.4 vs 2.3%)**.
- **Still: reduced movement warrants evaluation.** Low risk → fetal heart tones + adequate fluid may suffice; later gestation → **NST + amnionic fluid volume** (Parkland practice).
- **A single episode of excessive/vigorous movement** has been linked to later stillbirth — more research needed before acting on it.

2. Fetal Breathing

- **Discontinuous** and **paradoxical**: on inspiration the chest wall **collapses** and the abdomen protrudes (opposite of newborn/adult). Fluid exchange appears **essential for lung development**.
- Two types (Dawes): **gasps/sighs 1–4/min**, and **irregular bursts up to 240/min** (with REM).
- Respiratory rate ↓ while volume ↑ at **33–36 wk**, coinciding with lung maturation.

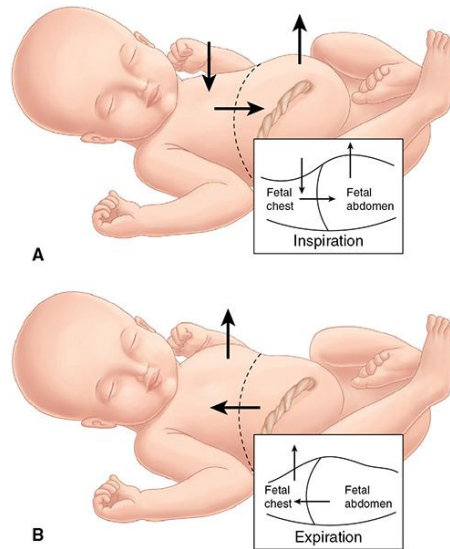


Figure 20-3. Paradoxical fetal breathing: chest collapses and abdomen protrudes on inspiration (A); chest expands on expiration (B).

- Affected by hypoxia and also: **maternal hypoglycemia**, sound, **smoking**, amniocentesis, **impending preterm labor**, gestational age, FHR, and **labor (breathing normally stops)**.
- **Diurnal**: less at night (minimum **20:00–24:00**), ↑ after meals, ↑ **04:00–07:00**.
- ⚠ Normal fetuses can have **no breathing for up to 122 min** → long observation needed. Not useful alone; part of the **BPP**.

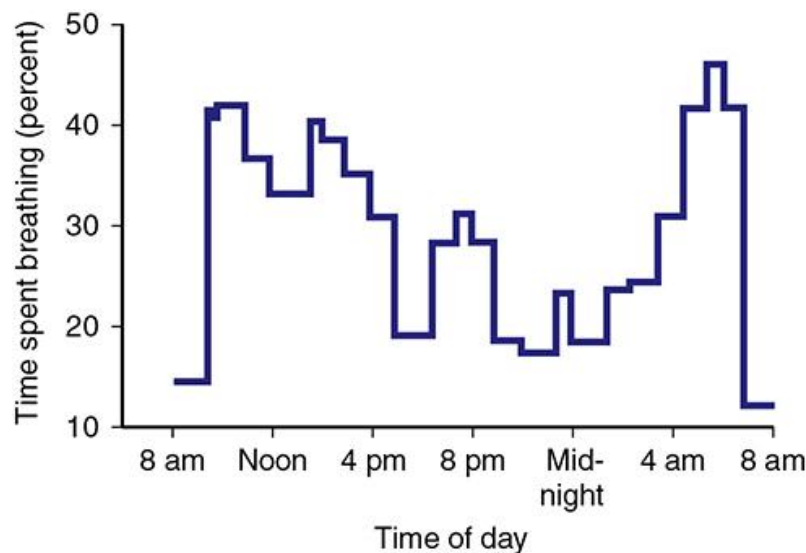


Figure 20-4. Time spent breathing at 38–39 wk: rises after breakfast, lowest 20:00–24:00, peaks 04:00–07:00.

3. Contraction Stress Test (CST)

- **Principle**: contractions compress myometrial vessels → ↓ intervillous flow. With **uteroplacental insufficiency** → **late decelerations** (start at contraction onset or after its acme). **Variable decelerations** suggest cord compression → often **oligohydramnios**.
- Originally the **oxytocin challenge test** (Ray, 1972). **Positive (abnormal)** = uniform repetitive **late decelerations**; negative = reassuring.
- CST = **test of uteroplacental function**.

Technique

- No stimulation needed if **≥3 spontaneous contractions ≥40 s in 10 min**.
- **Oxytocin** (Parkland): **20 U in 1 L** Ringer's, start **6 mU/min**, ↑ **6 mU/min every 40 min** to reach the pattern.

- **Nipple stimulation:** rub one nipple through clothing **2 min** or until a contraction → aim **3 per 10 min**; repeat after a **5-min rest**; then oxytocin if needed. Faster and cheaper than oxytocin; hyperstimulation reported by some.
- ⚠ Takes **~90 min** on average. **Relative contraindications = contraindications to vaginal delivery.**

Table 20-1. Criteria for Interpretation of the Contraction Stress Test

Result	Criteria
Negative	No late or significant variable decelerations
Positive	Late decelerations following $\geq 50\%$ of contractions (even if contraction frequency is fewer than three in 10 minutes)
Equivocal-suspicious	Intermittent late decelerations or significant variable decelerations
Equivocal-hyperstimulatory	FHR decelerations that occur with contractions more frequent than every 2 minutes or lasting longer than 90 seconds
Unsatisfactory	Fewer than three contractions in 10 minutes or an uninterpretable tracing

4. Nonstress Test (NST)

- Based on **FHR acceleration with fetal movement** in a fetus that is **not acidemic** or neurologically depressed. **Test of fetal condition** (vs CST = uteroplacental function).
- **Most widely used primary test**; easier than CST; part of the BPP. Accelerations diminish as hypoxia develops.

Accelerations (NICHD definitions)

Gestational age	Acceleration
≥ 32 wk	≥ 15 bpm above baseline for ≥ 15 s (and < 2 min)
< 32 wk	≥ 10 bpm for ≥ 10 s

- Percentage of movements with accelerations and their amplitude **rise with gestational age**. At **25–28 wk**, only **70%** reach 15 bpm but **90%** reach 10 bpm. RCT: 10×10 vs 15×15 criteria before 32 wk → same outcomes.

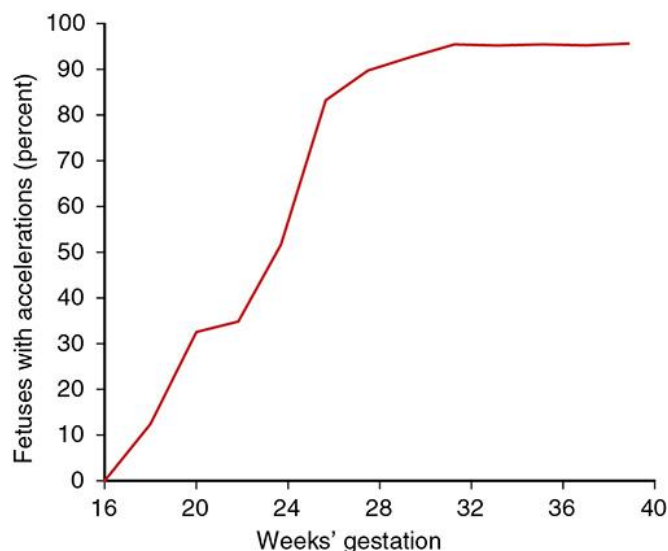


Figure 20-5. Fetuses with ≥ 1 15-bpm/15-s acceleration with movement: rises steeply from 24 wk to ~95% by 30–32 wk.

Reactive (normal) NST

- **≥ 2 accelerations within 20 min** of starting (with or without felt movement).
- **Run ≥ 40 min before calling it nonreactive** (sleep cycles). One acceleration may be as reliable as two.
- More often reactive, and faster, in the **evening**.

- **Vibroacoustic stimulation:** 1–2-s stimulus on the abdomen, repeatable **up to 3 times for up to 3 s** → shorter tests, fewer nonreactive results (one case of provoked tachyarrhythmia with known PACs).

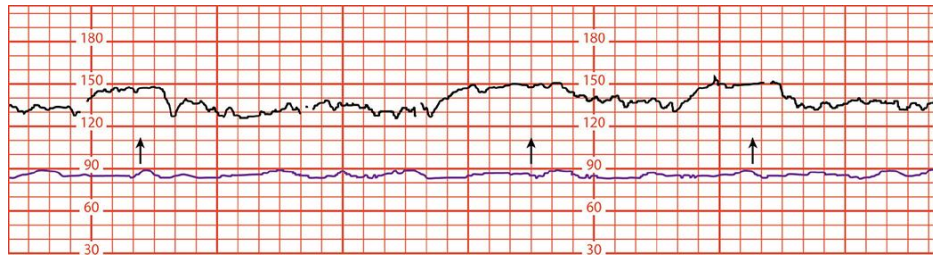


Figure 20-6. Reactive NST: at least two accelerations (arrows) of >15 bpm lasting >15 s.

Nonreactive (abnormal) NST

- Absence of accelerations **doesn't always mean compromise** — **false-positive rates ≥90%** reported; sleep is common.
- Healthy fetuses may not move for up to **75 min** → prolonged monitoring if nonreactive at **40 min**: becomes reactive within **80 min**, or stays nonreactive for **120 min** = very ill fetus.
- A nonreactive NST can **revert** when the fetal condition improves (e.g., after correcting maternal **DKA**), and a reactive one can become abnormal.

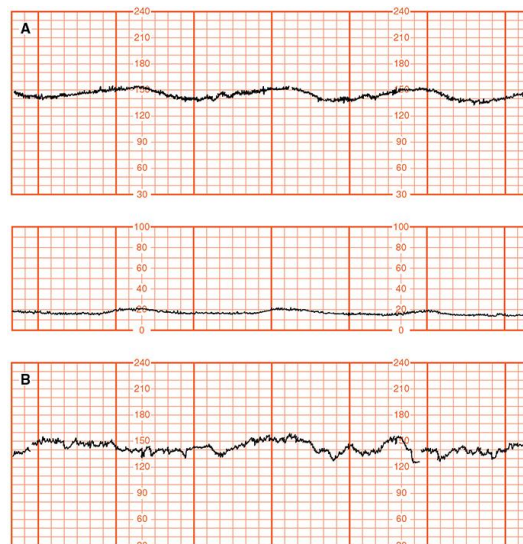


Figure 20-7. Maternal DKA at 28 wk: no accelerations, reduced variability and late decelerations (A) that normalize after acidemia is corrected (B).

Ominous patterns

- **Nonreactive ≥90 min** → **>90%** have significant perinatal pathology.
- **Terminal cardiotocogram (Visser):** (1) **baseline variability <5 bpm**, (2) **absent accelerations**, (3) **late decelerations with spontaneous contractions**.
- **Parkland:** **no accelerations over 80 min** (27 fetuses) → consistently uteroplacental pathology:

Finding	Frequency
Placental infarction	93%
Oligohydramnios	80%
Fetal-growth restriction	75%
Fetal acidemia	40%
Meconium	30%

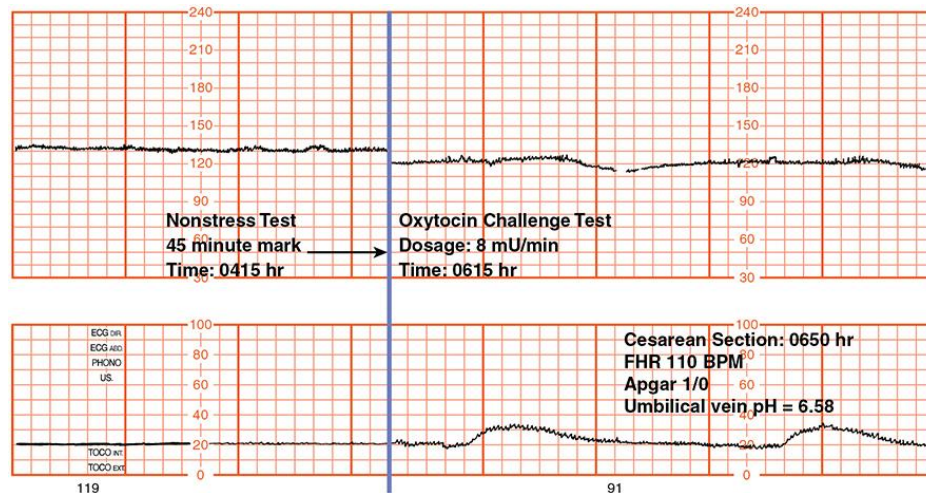


Figure 20-8. Nonreactive NST followed by a CST with mild late decelerations; at cesarean the fetus (umbilical vein pH 6.58) could not be resuscitated.

Interval between tests

- Originally (arbitrarily) **7 days**; varies by indication.
- **Weekly** for stable maternal disease (**pregestational diabetes, chronic hypertension, lupus**); **daily** by some for **preeclampsia remote from term**; individualized for fetal conditions (e.g., FGR).

Decelerations during NST

- Common with movement ($\frac{1}{2}$ – $\frac{2}{3}$ of tracings).
- **Nonrepetitive, brief (<30 s) variables** → **not compromise, no intervention needed.**
- **Repetitive variables (≥ 3 in 20 min)** → ↑ cesarean for fetal distress; decelerations ≥ 1 min → even worse prognosis.
- **Variables + oligohydramnios** → **75% cesarean rate**; distress also occurs with normal fluid.

False-reactive NST

- Fetal death within 7 days of a reactive NST: most common indication **postterm pregnancy**; mean interval **4 days** (1–7).
- Commonest autopsy finding **meconium aspiration**, often with a **cord abnormality** → acute asphyxia with gasping. **NST can't exclude acute asphyxia** → **assess amniotic fluid volume.**
- Other causes: **intrauterine infection, abnormal cord position, malformations, abruption.**

5. Biophysical Profile (BPP)

- **Five variables (Manning, 1980)**: NST, breathing, movement, tone, amniotic fluid. Each **2 (normal) or 0**; maximum **10**.
- Allow **30 min** before scoring breathing/movement as 0.
- Scores are higher in the **late evening**; **narcotics and sedatives lower** the score.

Table 20-2. Components and Scores for the Biophysical Profile

Component	Score 2	Score 0
Nonstress test ^a	≥2 accelerations within 20–40 min	0 or 1 acceleration within 20–40 min
Fetal breathing	≥1 episode of rhythmic breathing lasting ≥30 sec	<30 sec of breathing
Fetal movement	≥3 discrete body or limb movements	<3 discrete movements
Fetal tone	≥1 episode of extremity extension and subsequent return to flexion	0 extension/flexion events
Amnionic fluid volume ^b	A pocket of amnionic fluid that measures at least 2 cm in two planes perpendicular to each other (2 × 2 cm pocket)	Deepest single vertical pocket ≤2 cm

^aMay be omitted if all four sonographic components are normal. ^bFurther evaluation warranted, regardless of biophysical composite score, if deepest vertical amnionic fluid pocket ≤2 cm.

Memory hook: "2-3-1-30" — 2 accelerations, 3 movements, 1 flexion–extension, 30 s of breathing (plus a 2 × 2 cm pocket).

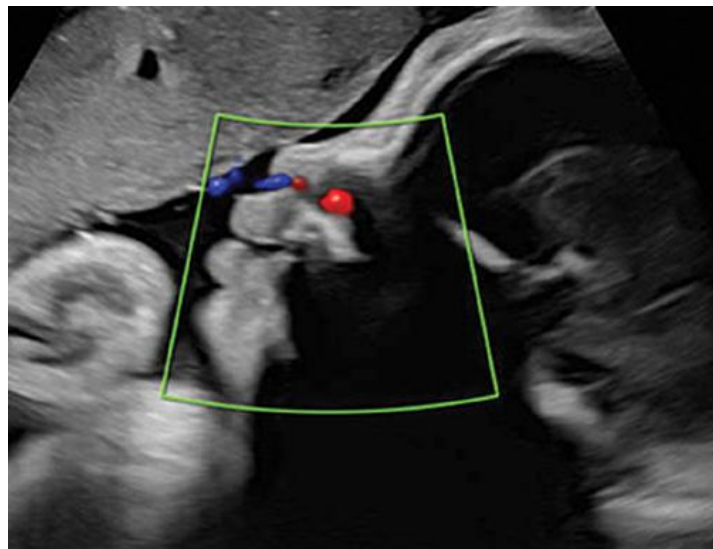


Figure 20-9. Color Doppler shows amnionic fluid flowing through the fetal nares with breathing.

Table 20-3. Interpretation of Biophysical Profile Score

Biophysical Profile Score	Interpretation	Recommended Management
10	Normal, nonasphyxiated fetus	No fetal indication for intervention; repeat test weekly
8/10 (Normal AFV) , 8/8 (NST not done)	Normal, nonasphyxiated fetus	No fetal indication for intervention; repeat test weekly
8/10 (Decreased AFV)	Chronic fetal asphyxia suspected	If ≥36 weeks, deliver. If <36 weeks, monitor per institution's protocol
6 (Normal AFV)	Equivocal	If ≥37 weeks, deliver. If <37 weeks and normal fluid, repeat test in 24 hours . If repeat test >6, monitor per institution's protocol
6 (Decreased AFV)	Possible fetal asphyxia	If 36–37 weeks, deliver. If <36 weeks, monitor per institution's protocol
4	Probable fetal asphyxia	If ≥32 weeks, deliver . If <32 weeks, individualize management based on maternal and fetal conditions
0 or 2	Almost certain fetal asphyxia	Deliver

AFV = amniotic fluid volume; NST = nonstress test.

- **Manning (>19,000 pregnancies): >97% normal; false-normal rate ~1 per 1000** (antepartum death of a normal fetus).
- Deaths after a normal BPP: **fetomaternal hemorrhage, cord accident, abruption.**
- **BPP vs umbilical venous pH (493 cordocenteses): score 0 → almost always acidemic; 8 or 10 → normal pH; 6 = poor predictor; a fall from 2–4 to 0 predicts better.** Overall **poor sensitivity** for cord pH.
- FGR fetuses **<1000 g** with daily BPPs: despite scores of 8 (27) or 6 (13), **6 died and 21 were acidemic.**

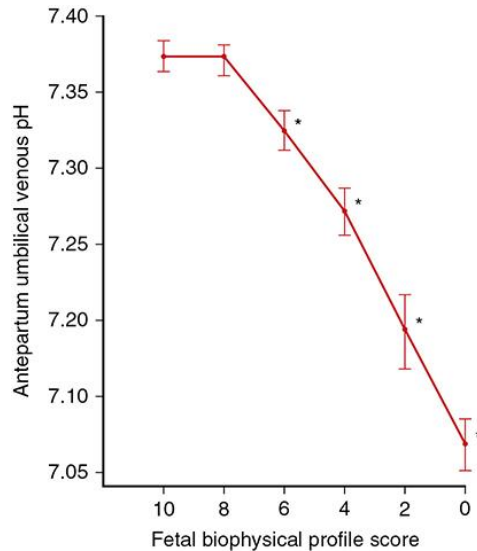


Figure 20-10. Mean umbilical venous pH falls as the BPP score falls (~7.37 at 8–10 to ~7.07 at 0).

Modified BPP

- **NST + amniotic fluid volume** (AFI; **≤5 cm abnormal**) — takes **~10 min**, no sonographer-heavy BPP.
- Clark (2628 pregnancies, **twice-weekly NST**): **no unexpected deaths.**
- Miller (>54,000 tests, 15,400 high-risk pregnancies): **false-negative 0.8 per 1000, false-positive 1.5%.**
- ACOG: **BPP and modified BPP are comparable** to other biophysical approaches.

6. Amniotic Fluid Volume

- Part of virtually every surveillance scheme. ↓ Uteroplacental perfusion → ↓ fetal renal flow → ↓ urine → **oligohydramnios.**
- Measured by **AFI** or **single deepest vertical pocket (DVP).**
- **Abnormal: AFI ≤5 cm; DVP ≤2 cm.**
- **AFI → more oligohydramnios diagnoses and inductions without better outcomes.** ACOG: **DVP preferred** (fewer unnecessary interventions, same outcomes).

7. Doppler Velocimetry

- Flow velocity reflects **downstream impedance.** Vessels: umbilical artery, MCA, ductus venosus, uterine artery.
- **Sequence of deterioration:** placental villous hypovascularity (**60–70% of small placental arteries obliterated** before the UA waveform is abnormal) → ↑ afterload (**>40% of combined ventricular output goes to the placenta**) → hypoxemia → ventricular dilation and **MCA redistribution** → ↑ **ductus venosus** pressure (right-heart afterload). **Abnormal DV = late finding.**

Umbilical artery

- **Abnormal S/D ratio: >95th percentile** for GA, or **absent/reversed end-diastolic flow** (poorly vascularized villi; extreme FGR; ↑ perinatal mortality).
- RCT (1360 high-risk): Doppler vs NST → more inductions but **cesarean for fetal distress 4.6% vs 8.7%**.
- BPP and UA Doppler are **comparable indicators of fetal acidosis**.
- ⚠️ **Not useful as a screening test in normal pregnancies. Valuable only in FGR (ACOG)** — best predictor of perinatal outcome in FGR.

Ductus venosus

- Abnormal waveform = **cardiac/myocardial dysfunction**, poorer outcome.
- **Not recommended for routine FGR surveillance** (SMFM).
- Used in **TTTS staging (Quintero)** and monitoring congenital heart defects and **SVT** (reversible cardiomyopathy → hydrops).

Middle cerebral artery

- **Primary method for detecting fetal anemia** (↑ **peak systolic velocity** from ↑ cardiac output and ↓ viscosity).
- **Not recommended for detecting fetal compromise**; adding MCA + UA Doppler to modified BPP did not change outcomes (RCT, 665 women).

Uterine artery

- Resistance normally ↓ in the first half of pregnancy (trophoblast remodeling).
- High resistance at **22–24 wk** linked to fetal death **<32 wk** with abruption, preeclampsia or FGR (30,519 women).
- **Not standard practice** in low- or high-risk populations (no technique standards or criteria).

Vessel	Main use	Recommended?
Umbilical artery	FGR surveillance	Yes, in FGR only
MCA	Fetal anemia (PSV)	Not for compromise
Ductus venosus	TTTS staging, cardiac/SVT monitoring	Not routine in FGR
Uterine artery	Uteroplacental insufficiency risk	Not standard

8. Antenatal Testing Summary

- Precision of every method is limited; testing forecasts **wellness** better than illness.
- **LA County (1971–85, ~17,000 women)**: testing ↑ from <1% to **15%** of pregnancies; fetal death rate lower in tested high-risk women (confounded by other innovations).
- Ghana (gestational hypertension): stillbirth **3.6 vs 9.2%** with NST (nonsignificant).
- **Cerebral palsy: 1.3 per 1000** in BPP-tested high-risk vs **4.7 per 1000** in untested low-risk pregnancies (Manning).
- By the time testing detects compromise, **damage may already be done** (Todd, Low).

Table 20-4. Indications for Antepartum Testing

Maternal	Pregnancy-related	Fetal
Chronic hypertension	Gestational hypertension	Fetal-growth restriction
Pregestational DM	Preeclampsia	Decreased fetal movement
SLE	Insulin-requiring gestational DM	Multifetal gestation
Antiphospholipid syndrome	Oligohydramnios	
Hemoglobinopathies	Polyhydramnios	
Cyanotic heart disease	Postterm pregnancy	
Cardiomyopathy	Prior stillbirth	
Cystic fibrosis	Isoimmunization	
Restrictive lung disease	Cholestasis	
Chronic renal disease	Velamentous cord insertion	
Hyperthyroidism	Single umbilical artery	
In vitro fertilization		
Substance abuse		
Chemotherapy (current)		
Prepregnancy BMI ≥ 35		
Maternal age >35		

BMI = body mass index; DM = diabetes mellitus; SLE = systemic lupus erythematosus.

Table 20-5. Stillbirth Rates within 1 Week of a Normal Antepartum Fetal Surveillance Test

Antepartum Fetal Test	Stillbirth ^a Rate/1000	Number
Nonstress test	1.9	5861
Contraction stress test	0.3	12,656
Biophysical profile	0.8	44,828
Modified biophysical profile	0.8	54,617

^aCorrected for lethal anomalies and unpredictable causes of fetal death such as abruption or cord accident.

When to start and how often

- Depends on **prognosis for neonatal survival** and severity of maternal disease.
- **Most high-risk pregnancies: start at 32–34 wk.** Severe complications (e.g., **FGR**): as early as **26–28 wk.**
- Repeat arbitrarily every **7 days** (often more frequently).

Quick-Recall: Antepartum Tests at a Glance

Test	Normal	Abnormal	Key fact
Fetal movement	Maternal impression (or 10 in 2 h)	Perceived decrease → NST + AFV	Informal = formal counting (Grant)
CST	Negative: no late/significant variables	Positive: lates after $\geq 50\%$ of contractions	Needs 3 contractions/10 min; ~ 90 min; stillbirth 0.3/1000
NST	Reactive: ≥ 2 accels in 20 min (15 $\times$ 15 at ≥ 32 wk; 10 $\times$ 10 < 32 wk)	Nonreactive after 40 min → extend; ≥ 90 min → $> 90\%$ pathology	Stillbirth 1.9/1000; FP $\geq 90\%$
BPP	8–10 (normal AFV)	≤ 4 probable asphyxia; 0–2 deliver	False-normal $\sim 1/1000$; score 0 → acidemia
Modified BPP	NST reactive + AFI > 5 cm	Nonreactive or AFI ≤ 5 cm	FN 0.8/1000, FP 1.5%
AFV	DVP > 2 cm, AFI > 5 cm	DVP ≤ 2 cm, AFI ≤ 5 cm	DVP preferred
UA Doppler	S/D ≤ 95 th percentile	S/D > 95 th; absent/reversed EDF	Only in FGR
MCA Doppler	—	$\uparrow$ PSV = anemia	Not for compromise

Reduces fetal movement / breathing	Increases activity	No effect
Smoking, methadone, buprenorphine, betamethasone (24–72 h) , narcotics/sedatives ($\downarrow$ BPP), oligohydramnios, labor (breathing stops), maternal hypoglycemia (breathing)	Glucose load , maternal meals (breathing), evening/night	Acetaminophen, maternal BMI (on perception)

Key Numbers to Memorize

- NPV $\geq 99.8\%$; PPV **10–40%**
- Movement from **7 wk**; behavioral states by **36 wk**; 1F + 2F = **75%** of time at 38 wk
- Sleep cycle **60 min** (quiet 23, active 40); longest normal inactivity **75 min**; apnea up to **122 min**
- Movements peak **~ 32 wk**; mothers perceive **16–45%**; decreased movement in **6–7%**
- CST: **≥ 3 contractions ≥ 40 s / 10 min**; oxytocin **6 mU/min $\uparrow$ 6 q40 min**; positive = lates with $\geq 50\%$; **~ 90 min**
- NST: **15 bpm $\times$ 15 s ≥ 32 wk, 10 $\times$ 10 < 32 wk**; reactive = **2 in 20 min**; run **40 min**; VAS up to **3 $\times$ 3 s**
- Terminal CTG: variability **< 5 bpm**, no accels, lates; no accels over 80 min → placental infarction **93%**
- Brief variables **< 30 s** OK; repetitive = **≥ 3 in 20 min**; ≥ 1 min → worse; variables + oligo → **75%** cesarean
- BPP: 30-min window; 2 $\times$ 2 cm pocket; false-normal **$\sim 1/1000$** ; score 4 → deliver ≥ 32 wk; 6 → repeat in 24 h if < 37 wk
- AFI **≤ 5 cm**, DVP **≤ 2 cm** abnormal
- UA Doppler abnormal after **60–70%** of placental small arteries obliterated; S/D **> 95 th percentile**
- Stillbirth ≤ 1 wk after normal test: NST **1.9**, CST **0.3**, BPP/modified BPP **0.8** per 1000
- Start testing at **32–34 wk** (FGR **26–28 wk**); repeat every **7 days**